YOUR MACHINES, IN FOCUS

Machine health overview.

Live telemetry, clear priorities. Stay one step ahead.

All systems within limits
System operating normally

MOTORS RUNNING

2 / 2 machines

Commanded PWM & direction

HIGHEST TEMPERATURE

29.1 °C

Warning >32° · Shutdown >35°

MOTION MAGNITUDE

2.1 °/s

Vibration 0.035 g · shared MPU6050

LOWEST REMAINING LIFE

10.00 hrs

10-hour initial usage budget / motor

Connected machines 02

Live digital view · commanded state

Drive motor

MG-001 / L298N · Channel A

RUNNING

TEMPERATURE	RISK SCORE	RUNTIME
28.4 °C	34 /100	0.00 h

Remaining usage budget 09:59:53 / 10h

Auxiliary motor

MG-002 / L298N · Channel B

RUNNING

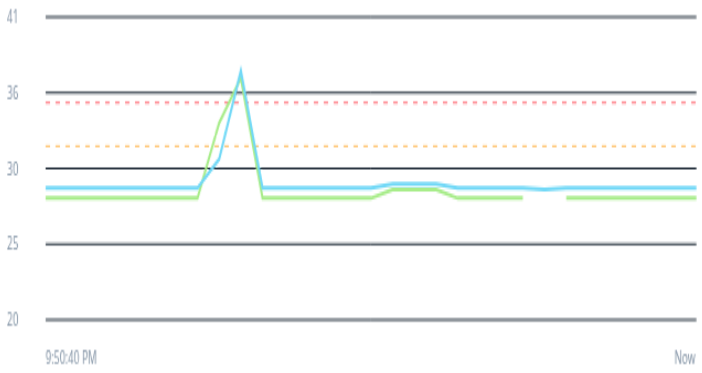
TEMPERATURE	RISK SCORE	RUNTIME
29.1 °C	41 /100	0.00 h

Remaining usage budget 09:59:54 / 10h

Sensor trends

Current session · last 180 samples

Motor 01 Motor 02 Dashed: warning / critical



Maintenance priority

RULE-BASED

01 **Auxiliary motor** 41
Routine monitoring · Routine lubrication / visual inspection

02 **Drive motor** 34
Routine monitoring · Routine lubrication / visual inspection

Risk is a transparent threshold score, not a trained failure prediction. Shared MPU motion applies to both motors. Remaining life counts powered running time; it is not a measured component lifetime.

↓ Draft maintenance work order

Failure classifier · research evaluation

Logistic regression · next-hour failure label · 20 held-out synthetic machines / 480 windows

SYNTHETIC DATA

ACCURACY	PRECISION / RECALL	F1 SCORE	ROC-AUC
84.79%	82.84 / 81.64%	0.8224	0.9179

Demonstration results only. Real-machine accuracy and remaining useful life have not been validated. This model does not control outputs; maintenance ranking uses the transparent threshold score above.

Controller activity

72 events · latest 32 retained

00:13:40	Command: M2F
00:13:40	Command: M1F
00:13:40	NORMAL: System operating normally
00:13:40	Command: RESET
00:13:40	Safety reset; outputs remain OFF
00:13:39	Simulator scenario: normal
00:13:39	CRITICAL: Sensor missing or stale
00:13:39	CRITICAL: Sensor missing or stale
00:13:39	Simulator scenario: disconnect
00:13:39	Command: M2F
00:13:39	NORMAL: System operating normally
00:13:39	Command: RESET
00:13:39	Safety reset; outputs remain OFF
00:13:37	Simulator scenario: normal
00:13:37	CRITICAL: Excessive movement / vibration
00:13:37	CRITICAL: Excessive movement / vibration
00:13:36	Simulator scenario: shock
00:13:36	WARNING: Sudden movement / vibration detected
00:13:36	Simulator scenario: motion
00:13:36	Command: M1F
00:13:35	NORMAL: System operating normally
00:13:35	Command: RESET
00:13:35	Safety reset; outputs remain OFF
00:13:33	Simulator scenario: normal
00:13:33	CRITICAL: Temperature above 35 C
00:13:33	CRITICAL: Temperature above 35 C
00:13:33	Simulator scenario: hot
00:13:32	WARNING: Temperature above 32 C

00:13:32	Simulator scenario: warm
00:13:32	Command: M2R
00:13:32	Command: M1F
00:13:32	NORMAL: System operating normally

Generated 10/28/2026, 9:50:52 PM · Source: simulator · Snapshot, not an equipment certification.